ADAPTIVE ENERGY CITY

A solar-aware energy scheduling simulator for buildings and urban districts

1. The Problem

Buildings use electricity at different times, while rooftop solar output changes with sunlight. Grid prices can rise during peak hours, and local feeders or transformers have capacity limits. If many loads run at once—or solar generation suddenly drops—buildings may draw more grid power than planned. Developers and solar providers need a way to test these situations before deployment.

2. The Proposed Solution

Create an interactive, top-down virtual city. Each building is a node with its own electricity demand, rooftop solar generation, flexible devices and connection limit. Animated connections represent the energy network. A scheduling engine detects shortages or overloaded connections, then shifts flexible activities—such as EV charging or water heating—to better times while keeping critical loads on.

3. How to Implement It

• Build a 2D city view with buildings, feeders and status colours.
• Simulate a 24-hour day in 15-minute steps.
• Give each building demand, solar output, tariffs, device priorities and deadlines.
• Add events: sudden solar drop, price spike and reduced grid capacity.
• Use Python to calculate demand and costs and reschedule flexible loads; use JavaScript Canvas for the animated map, with charts and a decision log.
• Compare the original and revised schedules using energy cost, peak demand, deferred load and deadline compliance.

4. What the Demo Shows

The user starts a normal day, triggers a solar drop or price spike, and watches the city update. Buildings and network links change status, flexible devices are rescheduled, and the system explains each decision. If no feasible schedule exists, it raises a warning instead of claiming the problem is solved.

5. Profitability Pitch

Potential customers include solar installation companies, building developers and energy-management providers. The product could be offered as:
• **Pre-sales planning:** demonstrate expected solar use and load-shifting opportunities to a developer.
• **Project assessment:** test different solar sizes, building loads and capacity limits before installation.
• **Software subscription:** provide scenario testing and reports for multiple projects.
• **Consulting or integration:** customize models using a customer's building and tariff data.

The business case is that customers may reduce avoidable peak-period energy costs, improve solar self-consumption and spot potential bottlenecks earlier. Revenue and savings must be validated with pilot projects; do not promise guaranteed savings or electrical safety from the simulation alone.

6. Scope and Limits

The first version models energy demand, solar generation, tariffs and feeder capacity. It is a planning and decision-support tool—not a certified electrical design or a guarantee of safe current flow. Real voltage, current, losses and protection studies would require a validated electrical power-flow model and engineering review.

Pitch: “We help solar companies and building developers test how buildings use solar power, manage flexible loads and respond to energy constraints—before committing to a real installation.”